



Unit 4 · Outcome 1 · Climate Change

Where does the Sun's energy go?

KK02

VCE Environmental Science · Units 3 & 4 · Exam study summary

Every second, Cape Wirreanda Climate Observatory is bathed in sunlight. Some of that energy bounces straight back to space off the snow on the ranges. Some is soaked up by the dark Southern Ocean. Some is radiated back out as heat — only to be caught by gases in the air and sent back down. This dot point is the story of those three fates: reflected, absorbed, and re-emitted.

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



THE BIG PICTURE

01 Earth runs on a solar budget

Think of incoming sunlight like income. Earth can't store sunlight forever — over the long run, the energy it takes in must balance the energy it sends back out to space. That balance is what sets the planet's temperature. Mess with either side and the temperature shifts.

The Sun sends energy as shortwave radiation — mostly visible light, high-energy, short wavelength. When sunlight reaches Earth, it meets one of three fates:

Bounced straight back to space by clouds, the atmosphere, and bright surfaces like ice, snow and desert. This is the albedo share — it never warms anything.

Soaked up by clouds, ozone, dust and water vapour in the atmosphere on the way down.

Soaked up by land and ocean, which warm and then re-emit the energy as longwave (infrared) heat.

◆ NOTE

○ Two kinds of radiation — don't mix them up Shortwave (incoming, from the Sun): high energy, short wavelength, visible light. Passes easily through the atmosphere. Longwave (outgoing, from Earth's warmed surface): lower energy, longer wavelength, infrared "heat". This is the radiation that greenhouse gases can absorb. The whole greenhouse story hinges on this difference.

◆ ANALOGY

Analogy — your bank account Sunlight is your pay coming in (shortwave). Spending is the heat radiated back to space (longwave). If income = spending, your balance (temperature) holds steady. Albedo is like income you never receive — reflected away before it can be "earned". The greenhouse effect is like a charge that delays your spending, so the balance sits higher than it otherwise would. Ocean circulation is the system that moves money between accounts so no single region goes broke or overflows.

REFLECTED ENERGY

02 The albedo effect

Albedo is the proportion of incoming shortwave (solar) radiation that a surface reflects, on a scale from 0 to 1 (or 0% to 100%). A perfect black absorber has an albedo of 0; a perfect white mirror has an albedo of 1. High-albedo surfaces stay cool because they bounce energy away; low-albedo surfaces warm up because they soak it in.

Surface	Typical albedo
Fresh snow	0.80–0.90
Sea ice	0.50–0.70
Desert sand	0.40
Grassland / crops	0.20–0.25
Eucalypt forest	0.08–0.12
Open ocean	0.05–0.06

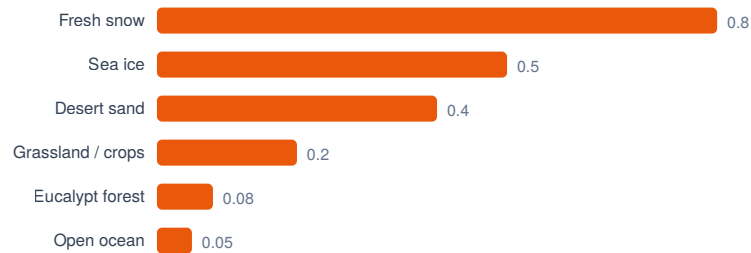


Figure - Surface compared at a glance.

Earth's overall average albedo is about 0.30 — roughly 30% of all incoming sunlight is reflected before it can warm the planet.

◆ KEY INSIGHT

The exam-critical insight At Cape Wirreanda, the snow-capped ranges reflect most of their sunlight (high albedo, stays cold) while the dark Southern Ocean reflects almost none (low albedo, warms up). Change the surface, and you change how much energy Earth keeps. That single sentence is the heart of every albedo question.

INTERACTIVE · ICE-ALBEDO FEEDBACK

03 Albedo lab: melt the ice, feel the warming

◆ ANALOGY

Analogy — black vs white t-shirt On a hot January day you wear white, not black. A white shirt has high albedo and reflects the Sun, so you stay cooler; a black shirt has low albedo, soaks up the light, and bakes you. Earth's icecaps are the planet's white shirt. Melt them, and the dark ocean underneath is the black shirt — and the planet runs warmer.

⚠ WATCH OUT

Why this is a feedback loop Warming → ice melts → albedo drops → more sunlight absorbed → more warming → more ice melts... The output (warming) feeds back to amplify the cause. That's a positive feedback — "positive" means amplifying, not "good". This is exactly why the poles are warming faster than anywhere else on Earth.

RE-EMITTED ENERGY

04 The natural greenhouse effect

Here's the puzzle: if shortwave sunlight passes straight through the atmosphere, why doesn't all the heat just escape straight back out again? The answer is the change in wavelength when energy is re-emitted.

- Shortwave in: Sunlight passes through the atmosphere (greenhouse gases are transparent to it) and is absorbed by the land and ocean.
- Surface warms and re-emits the energy — but now as longwave (infrared) radiation, because the surface is far cooler than the Sun.
- Greenhouse gases absorb this outgoing longwave radiation. Water vapour, carbon dioxide, methane, nitrous oxide and ozone are all "longwave-active".

- They re-emit it in all directions — including back down toward the surface (this is called back-radiation). That returned energy warms the lower atmosphere and surface a second time.

◆ KEY INSIGHT

The number that proves it matters Without the natural greenhouse effect, Earth's average surface temperature would be about -18°C — a frozen rock. With it, the average is a comfortable $+15^{\circ}\text{C}$. That's roughly 33°C of natural warming that makes life possible. The natural greenhouse effect isn't the problem — it's the reason we're here. (The enhanced greenhouse effect, where humans add extra gas, is the next dot point.)

◆ ANALOGY

Analogy — a doona on a cold night Greenhouse gases work like a doona (blanket). The doona doesn't make heat — your body does. But it traps the heat radiating off you and keeps you warm. Pile on a thicker doona (more greenhouse gas) and you get warmer still. A clear, doona-less winter night radiates heat straight to the sky and you freeze — just like Earth would at -18°C . (Note: a real greenhouse warms mostly by blocking convection, not by this radiation trick — but the "blanket that traps escaping heat" image is the right intuition for the atmosphere.)

MOVING THE ENERGY AROUND

06 Ocean circulation: the planet's heat conveyor

Albedo and the greenhouse effect decide how much solar energy Earth keeps. Ocean circulation decides where it goes. Oceans absorb a huge share of incoming sunlight and — because water holds heat far better than air or land — they act as a vast thermal store, then carry that heat thousands of kilometres around the globe.

Driven by prevailing winds. Off Cape Wirreanda, the warm East Australian Current carries tropical heat south down the coast (yes — the "EAC" from Finding Nemo), moderating the climate of south-eastern Australia.

The deep "global conveyor belt", driven by differences in temperature (thermo) and salinity (haline). Cold, salty, dense water sinks near the poles; warm surface water flows in to replace it. A single loop takes ~ 1000 years.

◆ NOTE

≈ How sinking water drives the whole belt Near Antarctica and the North Atlantic, surface water is cold and, as sea ice forms, the leftover water becomes saltier. Cold + salty = dense, so it sinks. This sinking pulls warm surface water poleward to replace it, releasing its stored heat to the air on the way. The result is a planet-spanning loop that carries equatorial heat toward the poles and keeps regional climates far milder than they'd otherwise be.

◆ ANALOGY

Analogy — central heating for the planet Ocean circulation is Earth's central-heating system. The equator is the boiler, soaking up the most intense sunlight. The currents are the pipes, pumping warm water out to the cold "rooms" near the poles and returning cool water to be reheated. Without the pump, the tropics would overheat and the poles would freeze solid. The ocean's huge heat capacity also makes it a flywheel — it stores energy and releases it slowly, smoothing out what would otherwise be wild swings between day and night, summer and winter.

SYNTHESIS

07 The three working together at Cape Wirreanda

A single sunbeam arriving at the observatory tells the whole story:

If it lands on snow it's reflected and lost. If it lands on dark ocean it's absorbed and kept. Albedo sets how much energy stays.

The absorbed energy is re-emitted as longwave, then trapped and recycled by greenhouse gases, raising the surface temperature $\sim 33^\circ\text{C}$ above frozen.

Currents carry the kept heat around the planet, storing it and spreading it from equator to poles so no region runs to extremes.

Together they explain why Earth sits at a liveable average of $\sim 15^\circ\text{C}$ with surprisingly mild regional climates — and why a change to any one of them (melting ice, more gas, a slowing conveyor) ripples through the entire system.

DON'T LOSE EASY MARKS

08 Six exam traps

⚠ WATCH OUT

Trap 1 — "Albedo is heat / temperature" Albedo is reflectivity — the proportion of sunlight reflected. It is not a temperature or an amount of heat. Define it as a fraction or percentage from 0 to 1.

⚠ WATCH OUT

Trap 2 — "Greenhouse gases trap incoming sunlight" They don't. Greenhouse gases are transparent to incoming shortwave. They absorb the outgoing longwave (infrared) the surface re-emits. The wavelength change is the whole mechanism — say it explicitly.

⚠ WATCH OUT

Trap 3 — Confusing the natural and enhanced greenhouse effects This dot point is about the natural effect — the baseline that keeps Earth at $+15^\circ\text{C}$ and is essential for life. Don't slip into talking about human emissions and global warming unless the question asks for it.

⚠ WATCH OUT

Trap 4 — "Positive feedback is a good thing" "Positive" means amplifying, not beneficial. The ice–albedo loop is a positive feedback that makes warming worse.

⚠ WATCH OUT

Trap 5 — Forgetting back-radiation Greenhouse gases re-emit absorbed energy in all directions, including back toward the surface. The downward (back) radiation is what does the warming. "Absorb and re-emit" earns more than just "absorb".

⚠ WATCH OUT

Trap 6 — Treating the ocean as just a passive sponge Oceans don't only store heat — they move it. A complete answer mentions redistribution by currents (surface and thermohaline), not just absorption.

09 Command-word decoder

◆ COMMAND WORDS

The verb in the question tells you how much to write. Tap each to see what the examiner actually wants.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

BUILD A TOP-BAND ANSWER

11 P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

VCE markers reward structure. The P·E·L method turns a vague paragraph into a mark-grabbing one. Use it on every "explain" question. Question: Explain how the albedo effect influences the amount of solar energy Earth retains. (3 marks) State the relationship up front: albedo is the proportion of sunlight reflected, and it sets how much energy is kept vs lost. Contrast a high-albedo and a low-albedo surface with figures (snow ~0.8 vs ocean ~0.06) and say what each does to absorbed energy. Connect it to the consequence: more reflective surface → less energy absorbed → cooler; and flag the feedback if melting changes the surface. Write your 3-mark answer, then reveal a model response:

SEE THE DIFFERENCE

12 Weak vs strong: a worked answer

Question: Explain the natural greenhouse effect and why it is important for life on Earth. (4 marks)

Greenhouse gases trap the sun's heat in the atmosphere like a greenhouse. This warms up the Earth. Without it the Earth would be cold. Gases like CO₂ cause the greenhouse effect.

- **NOTE**

Vague and mechanism-free. "Trap the sun's heat" misses the shortwave-in / longwave-out distinction — the actual mechanism. No mention of absorption and re-emission, no back-radiation, no figures. "Earth would be cold" isn't specific enough to score. Earns 1 mark for the basic idea that gases retain heat and warm Earth.

Incoming shortwave solar radiation passes through the atmosphere and is absorbed by Earth's surface, which warms and re-emits the energy as longwave (infrared) radiation. Greenhouse gases such as water vapour and CO₂ absorb this outgoing longwave radiation and re-emit it in all directions, including back toward the surface, retaining heat in the lower atmosphere. This keeps Earth's average temperature around +15 °C instead of about -18 °C, making the planet warm enough to support liquid water and life.

- **NOTE**

Four distinct marks: (1) shortwave absorbed by surface; (2) re-emitted as longwave; (3) GHGs absorb and re-emit/back-radiate; (4) quantified importance for life. Each highlighted clause is a scoring point.

EXAM-STYLE · 20 MARKS

13 Practice exam questions

- ★ **PRACTICE & SELF-MARK**

Five VCAA-style questions. Try each one first, then reveal the marking guide and a weak vs strong sample. Mark yourself honestly — your score builds in the dock at the bottom right. Define albedo and explain how the albedo of a surface affects the amount of solar energy that surface absorbs.

- 1 mark — Albedo = the proportion (fraction/percentage, 0–1) of incoming solar (shortwave) radiation that a surface reflects.
- 1 mark — A higher albedo means more sunlight is reflected and less absorbed (cooler); a lower albedo means less is reflected and more is absorbed (warmer).

- ★ **PRACTICE & SELF-MARK**

Albedo is how shiny something is. Shiny things don't get as hot.

- **NOTE**

"How shiny" is too loose and never defines albedo as a proportion of reflected solar radiation. No clear link to absorbed energy.

- ★ **PRACTICE & SELF-MARK**

Albedo is the proportion of incoming solar radiation that a surface reflects, on a scale from 0 to 1. A surface with a high albedo (e.g. snow, ~0.8) reflects most sunlight and absorbs little, so it stays cool, whereas a low-albedo surface (e.g. ocean, ~0.06) reflects little and absorbs most of the incoming energy.

- **NOTE**

Definition as a proportion + the absorb/reflect relationship with an example. Full marks.

★ PRACTICE & SELF-MARK

Describe how the natural greenhouse effect retains heat in Earth's atmosphere. Refer to both shortwave and longwave radiation in your answer.

- ① Incoming shortwave solar radiation passes through the atmosphere and is absorbed by Earth's surface.
- ② The warmed surface re-emits the energy as longwave (infrared) radiation.
- ③ Greenhouse gases absorb this outgoing longwave radiation.
- ④ They re-emit it in all directions, including back toward the surface (back-radiation), retaining heat.

★ PRACTICE & SELF-MARK

The sun's rays come in and the greenhouse gases trap them in the atmosphere so the heat can't get out, which warms the Earth up.

• NOTE

"Trap the rays" conflates shortwave and longwave and misses that the radiation is re-emitted at a longer wavelength. No back-radiation. Scores the entry-level idea only.

★ PRACTICE & SELF-MARK

Shortwave solar radiation passes through the atmosphere and is absorbed by the surface, which warms and re-emits the energy as longwave infrared radiation. Greenhouse gases such as CO₂ and water vapour absorb this outgoing longwave radiation and re-emit it in all directions, including back down to the surface, which retains heat in the lower atmosphere.

• NOTE

All four steps, with the shortwave→longwave conversion and back-radiation explicit.

★ PRACTICE & SELF-MARK

As polar sea ice melts, scientists observe that warming in polar regions accelerates. Explain this observation using the concept of albedo, and identify the type of feedback involved.

- 1 mark — Sea ice has a high albedo and reflects most incoming sunlight.
- 1 mark — When ice melts, it exposes darker ocean with a much lower albedo.
- 1 mark — The darker ocean absorbs more solar radiation, causing further warming and more melting.
- 1 mark — This is a positive (amplifying) feedback loop.

★ PRACTICE & SELF-MARK

The ice melts because it gets hot and then the water is dark so it gets hotter. It's a feedback loop.

• NOTE

Right intuition but no albedo terminology, no high-vs-low contrast, and "feedback loop" isn't named as positive/amplifying. Loosely worded.

★ **PRACTICE & SELF-MARK**

Sea ice has a high albedo, reflecting most incoming solar radiation. As it melts it exposes darker ocean water, which has a much lower albedo and absorbs far more of that radiation. The extra absorbed energy raises temperatures and melts more ice, exposing still more dark ocean — so the warming amplifies itself. This is a positive feedback loop.

• **NOTE**

High→low albedo contrast, the absorption mechanism, the self-reinforcing chain, and the correct label "positive feedback".

★ **PRACTICE & SELF-MARK**

Explain how ocean circulation influences the distribution of solar energy across Earth. Refer to both surface currents and thermohaline circulation.

- 1 mark — Oceans absorb a large proportion of incoming solar energy and store it (high heat capacity).
- 1 mark — Wind-driven surface currents (e.g. the East Australian Current) carry warm water from the tropics toward higher latitudes.
- 1 mark — Thermohaline circulation is driven by differences in temperature and salinity (density); cold, salty water sinks at the poles.
- 1 mark — This redistributes heat from the equator toward the poles, moderating regional climates.

★ **PRACTICE & SELF-MARK**

The ocean moves heat around in currents and that warms up some places. It also has deep currents.

• **NOTE**

Names "currents" but doesn't distinguish surface from thermohaline, gives no driver (wind / temperature + salinity), and no equator-to-pole redistribution.

★ **PRACTICE & SELF-MARK**

Oceans absorb much of the incoming solar energy and store it because water has a high heat capacity. Wind-driven surface currents, such as the East Australian Current, carry warm tropical water poleward. Deeper thermohaline circulation is driven by differences in temperature and salinity: cold, salty, dense water sinks near the poles, pulling warm water in to replace it. Together these move heat from the equator toward the poles, moderating climates that would otherwise be far more extreme.

• **NOTE**

Storage + both current types with their drivers + equator-to-pole redistribution. Complete.

★ PRACTICE & SELF-MARK

A sunbeam strikes the dark ocean off Cape Wirreanda at midday. Trace what happens to that energy, referring to all three of the albedo effect, the natural greenhouse effect and ocean circulation. Explain how, together, these processes help keep Earth's climate liveable.

- Albedo (1–2) — Dark ocean has a low albedo (~ 0.06), so it reflects little and absorbs most of the shortwave radiation; a high-albedo surface (snow/ice) would have reflected it instead.
- Greenhouse (1–2) — The warmed surface re-emits energy as longwave radiation; greenhouse gases absorb and re-emit it (including back-radiation), retaining heat and keeping Earth $\sim 33^\circ\text{C}$ warmer ($-18 \rightarrow +15^\circ\text{C}$).
- Ocean circulation (1–2) — The absorbed heat is stored by the ocean and redistributed by surface and thermohaline currents from equator toward poles, moderating regional extremes.
- Synthesis (1) — Together: albedo sets how much energy is kept, the greenhouse effect retains and warms, and circulation spreads it — producing a stable, liveable average climate.

★ PRACTICE & SELF-MARK

The sun hits the dark ocean so it gets absorbed because dark things absorb heat. Then the greenhouse gases trap the heat and it warms the Earth. The ocean currents move the heat around to different places. All of this keeps the Earth warm enough to live on.

• NOTE

Touches all three but with no figures, no shortwave→longwave conversion, no albedo value, no current types, no back-radiation. Each idea is asserted, not explained — caps in the lower band.

★ PRACTICE & SELF-MARK

Because the ocean is dark it has a low albedo (~ 0.06), so it reflects very little of the incoming shortwave radiation and absorbs almost all of it — unlike high-albedo snow, which would have reflected most of it back to space. The warmed water re-emits this energy as longwave infrared radiation, which greenhouse gases absorb and re-emit in all directions, including back to the surface; this back-radiation keeps Earth around $+15^\circ\text{C}$ rather than -18°C . The ocean stores this heat and redistributes it through wind-driven surface currents and density-driven thermohaline circulation, carrying warmth from the equator toward the poles. Together, albedo controls how much energy is retained, the greenhouse effect traps and recycles it, and ocean circulation spreads it evenly — producing the stable, moderate climate that makes life possible.

• NOTE

Albedo value + shortwave/longwave + back-radiation + temperature figures + both current types + a clear synthesis sentence. Top band.

◆ KEY INSIGHT

Before you move on to KK03 You should now be able to: define albedo as a proportion of reflected sunlight and give surface values; describe the natural greenhouse effect using the shortwave→longwave→back-radiation chain and the $-18 \rightarrow +15^\circ\text{C}$ figures; explain the ice–albedo positive feedback; and describe how surface and thermohaline ocean currents redistribute absorbed solar energy. Next dot point (KK03) follows the energy once it's absorbed — into carbon sequestration and the carbon cycle.

Where does the Sun's energy go?

Every second, Cape Wirreanda Climate Observatory is bathed in sunlight. Some of that energy bounces straight back to space off the snow on the ranges. Some is soaked up by the dark Southern Ocean. Some is radiated back out as heat — only to be caught by gases in the air and sent back down. This dot point is the story of those three fates: reflected, absorbed, and re-emitted.

This dot point asks you to describe the interactions of solar energy absorbed, re-emitted and reflected by atmospheric gases and matter, including the albedo effect, the natural greenhouse effect and ocean circulation. By the end of this module you'll be able to explain all three — and tell an examiner exactly how they keep Earth liveable.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Greenhouse gases trap the sun's heat in the atmosphere like a greenhouse. This warms up the Earth. Without it the Earth would be cold. Gases like CO₂ cause the greenhouse effect. Vague and mechanism-free. "Trap the sun's heat" misses the shortwave-in / longwave-out distinction — the actual mechanism. No mention of absorption and re-emission, no back-radiation, no figures. "Earth would be cold" isn't specific enough to score. Earns 1 mark for the basic idea that gases retain heat and warm Earth.

✓ STRONG / MODEL ANSWER

Incoming shortwave solar radiation passes through the atmosphere and is absorbed by Earth's surface, which warms and re-emits the energy as longwave (infrared) radiation. Greenhouse gases such as water vapour and CO₂ absorb this outgoing longwave radiation and re-emit it in all directions, including back toward the surface, retaining heat in the lower atmosphere. This keeps Earth's average temperature around +15 °C instead of about –18 °C, making the planet warm enough to support liquid water and life. Four distinct marks: (1) shortwave absorbed by surface; (2) re-emitted as longwave; (3) GHGs absorb and re-emit/back-radiate; (4) quantified importance for life. Each hi

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✓ STRONG / MODEL ANSWER

Albedo is the proportion of incoming solar radiation that a surface reflects, on a scale from 0 to 1. A surface with a high albedo (e.g. snow, ~0.8) reflects most sunlight and absorbs little, so it stays cool, whereas a low-albedo surface (e.g. ocean, ~0.06) reflects little and absorbs most of the incoming energy. Definition as a proportion + the absorb/reflect relationship with an example. Full marks.

✗ WEAK ANSWER

The sun's rays come in and the greenhouse gases trap them in the atmosphere so the heat can't get out, which warms the Earth up. "Trap the rays" conflates shortwave and longwave and misses that the radiation is re-emitted at a longer wavelength. No back-radiation. Scores the entry-level idea only.

✓ STRONG / MODEL ANSWER

Shortwave solar radiation passes through the atmosphere and is absorbed by the surface, which warms and re-emits the energy as longwave infrared radiation. Greenhouse gases such as CO₂ and water vapour absorb this outgoing longwave radiation and re-emit it in all directions, including back down to the surface, which retains heat in the lower atmosphere. All four steps, with the shortwave→longwave conversion and back-radiation explicit.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.